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RESEARCH PAPER

Enculturation modulates aesthetic judgments of dance

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Abstract Seasonal and other festivities with dance and music shape individuals' aesthetic sensitivity to their cultural heritage. Yet, there is little empirical research about the influence of culture on dance perception. To fill the gap, this paper presents a pre-registered experiment with 120 participants (Iranian, English, Southeast Asian), that investigated the influence of *cultural group* (categorical variable) and *enculturation* (continuous variables) on aesthetic judgment to emotionally expressive Iranian classical dance stimuli. We also explored how culture and enculturation modulated responses to the question 'how important is the cultural heritage of 'Persian dance' for you?'. Education, age, motion energy, and aesthetic responsiveness served as covariates across all models. Our data served to explore an important question in cross-cultural empirical aesthetics: which type of variable best predicts the influence of culture on questions about a cultural heritage like dance—categorical or continuous culture variables? Surprisingly, there were no overall differences in aesthetic judgment between the groups, although Iranians and English participants were most sensitive to joyful emotional expressivity of the dancer. Aesthetic judgment was modulated by English, but not Iranian enculturation. Conversely, Iranians found Persian

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dance more important than did the other two groups. In all analyses, aesthetic responsiveness was a significant covariate, while education, motion energy and age were not. Effect sizes (partial η^2s and βs) were higher for continuous enculturation measures, than for the categorical variable *cultural group*. We recommend using continuous enculturation variables in cross-cultural empirical aesthetics research to allow higher granularity when exploring the impact of culture on perception.

Keywords Aesthetics · Dance · Enculturation · Cross-cultural · Full-body movement · Non-verbal communication · Expressivity · Iran · Europe · Persia · Dance · Iranian dance · Emotion

Introduction

Empirical Aesthetics, a subdiscipline of experimental psychology (Wassiliwizky & Menninghaus, 2021), often focuses on the relation between perceptual properties of stimuli (e.g., expressivity, emotion, colour, shape, etc.), and observers' aesthetic response (Chatterjee, 2011). Relatively recently, dance as an artform has joined visual art and music as a subject for empirical aesthetics (Christensen & Calvo-Merino, 2013; Christensen & Jola, 2015; Cross, 2025; Cross, 2011; Meletaki et al., *in review*).

Through studies from different domains of the arts, we know that aesthetic experiences can be strongly dependent on the culture or origin of the participants (Bao et al., 2016; Darda & Chatterjee, 2023; Darda & Cross, 2022; Jacoby et al., 2020; Masuda et al., 2008; Nisbett & Miyamoto, 2005; Vessel et al., 2018; Wang et al., 2012). However, there is still little empirical research about the influence of culture on dance perception, and this paper aims to fill this gap. Iranian dances are fairly unknown in the West. Hence, comparing Iranian and Western participants' aesthetic judgments to Iranian dance stimuli is a rather unique opportunity to investigate the influence of enculturation on aesthetic judgment.

'Iranian dances' refers to a rather heterogenous group of dances that are practiced within the geographical boundaries of Iran today. These dance styles include folk dances practiced by various tribes in the rural regions (e.g., *Balouchi, Kurdish, Bojnurdi, Bandari, Turkmen*, etc.; Gholami, 2016), popular social dance, which is an urban dance style but practiced nationwide, and the staged version of urban dance, sometimes referred to as "Iranian classical dance" (Christensen et al., 2024a, 2024b). Therefore, a short clarification is in order. For the present investigation, we relied on one specific Iranian dance style that the team has already used in prior work. The nomenclature of how to refer to the dance style is contested, and the discussion lies outside the scope of this paper[1]. For the sake of consistency with our previous work, we will refer to it here as 'Iranian classical dance'. The reason is that—similar to Western Ballet and Indian Bharatanatyam—this dance style likely originated in royal courts. There is some repertoire and archive, and while the vernacular version of it is practiced in urban, social dance settings, the staged, concert-style version of this genre is practiced and performed in the underground sectors of Iran, and in many parts of the world (Christensen et al., 2024a, 2024b). However, we point out that the same dance style may also be referred to as 'Iranian urban dance' (Ameri, 2006; Khorsandi, 2026),'Persian dance' (Friend, 1996), and others (Christensen et al., 2024a, 2024b; Khorsandi, 2015; Shay, 1998; Shay, 1999). Khorsandi et al. (*under review*) created a stimuli library of 100 Iranian classical dance sequences (6s length), which express joy, sadness, anger, fear, and neutrality. A validation experiment (Christensen et al., 2025), confirmed that both Iranian and English participants were able to recognize these emotional expressivities, intended by the dancer while dancing, above chance level. Iranian participants' aesthetic judgments of the beauty and likability of the full set of these stimuli were higher than English participants' (Khorsandi et al., *under review*), especially to stimuli, where the dancer intended to express joy. Stylistic norms on dance expressivities in Iran hold that joy is the natural expressivity of this dance style, which means that the results in this previous

[1] For further discussions about how to refer to the dances of Iran, please see other work by Shahrzad Khorsandi and Anthony Shay. For example: Khorsandi and Christensen (2026) Iranian Dancers Perform Despite Restrictions: Navigating a Sociopolitical Obstacle Course In A. Shay (Ed.), *Middle Eastern Dance: A Reader*. Palgrave Macmillan. Or: Shay (1999). *Choreophobia—Iranian Solo Improvised Dance*. Mazda Publishers, U.S.

experiment evidenced an effect of culture on aesthetic judgment.

For the present investigation, the 25 stimuli associated with the highest recognition rates in the validation study (Christensen et al., 2025) were used, five stimuli for each of the five emotional expressions. We predicted that joyful stimuli would receive the highest aesthetic judgments, thus replicating the findings reported by Khorsandi, et al. (*under review*). However, we also aimed to provide a more fine-grained investigation of the effect of culture on aesthetic judgment to these Iranian classical dance stimuli.

In 2010, Henrich et al. famously called for more research with participants who were not from societies described as Western, educated, industrialized, rich, and democratic (Henrich et al., 2010). The objective of Henrich et al. (2010) was to promote research about the human psyche in general, and not to limit the scientific record to studies of Western samples. The field has grown since then (Apicella et al., 2020). One interesting phenomenon that has emerged is that, with societies becoming increasingly interconnected through the internet, social media, travel and migration, a categorical separation of research participants into 'cultural groups' (meaning their self-ascribed culture of origin, their legal nationality, country of bith, country of residence, ethnicity, language group or other 'culture group' identification like minority culture group membership) may no longer be optimal. For example, Yang et al. (2019) found that participants who had self-identified as Chinese, but lived in the US, responded to an aesthetic judgment task in a way that was much more similar to US-American participants' responses, than to aesthetic judgments by Chinese participants who lived in China (Yang et al., 2019). This finding was no exception (Bao et al., 2016; Masuda et al., 2008), leading to the suggestion that instead of a categorical 'culture group' variable, continuous measurements of research participants' *enculturation* with a given culture may offer a more granular perspective on the influence of culture on behaviour (Christensen et al., 2024a, 2024b). Enculturation is the neurodevelopmental process of growing up in a particular cultural community and being exposed to its specific socio-cultural learning environment. Some work in immigration psychology (Curran, 2012; Kirshner & Meng, 2012) and in empirical aesthetics (for a review see Jacobsen & Beudt, 2017), suggests that the first years of life up until early adolescence (10–12 years of age) are particularly formative and are considered important for enculturation processes to take place.

Three culture screening measures that return continuous variables of enculturation were developed by an international research team (some of whom are also part of the present author team; Christensen et al., 2024a, 2024b). The conceptualization of these measures was based on questionnaire development in immigration psychology (for instance, see the review by Kirshner & Meng, 2012), specifically on the Frankfurt Acculturation Scale by Frankenberg & Bongard (2013) (Frankenberg & Bongard, 2013; Kirshner & Meng, 2012). Enculturation to a given cultural sphere has been shown to shape peoples' perceptions and feelings of connectedness to traditions and arts of that culture (Bornstein & Cote, 2004; Farver & Lee-Shin, 2000; Frankenberg & Bongard, 2013; Golbabaei et al., 2022; Hall, 1966; Hannon, (2010); Kirschner & Martin, 2010; Miyawaki, 2015; Polak et al., 2026; Rudmin, 2003; Tsai & Chentsova-Dutton, 2002; Ying et al., 2000; Yu et al., 2016). For example, seasonal events and other festivities, where cultural traditions and arts are practiced, offer rich opportunities to shape people's prior experiences with a culture's art, foods, music, and dance (Christensen et al., 2024a, 2024b; Polak, 2019).

We therefore recruited Iranian and English participants for our study via the online survey tool Prolific® (Peer, et al. 2021) by using several filters provided on the platform for participant selection (e.g., nationality, first language, etc.—more details in the methods section). Additionally, we also administered the three newly developed screening measures of enculturation to explore our participants' Iranian and English enculturation (Christensen et al., 2024a, 2024b). To increase the sensitivity of our analyses, we added another cultural group: Southeast Asian participants who had been born and raised in England (for this group, 'culture group' was selected again by setting the filters on the Prolific platform to the relevant 'nationality', 'language', 'country of birth', 'first language', etc.—more information in the methods section). These participants were thought to be enculturated with English culture to a lesser degree than the English participants, but more so than the Iranians. Similarly, we expected them to be less enculturated with Iranian traditions than the Iranians and the English participants.

Summing up, we investigated the influence of *cultural group* and *enculturation* on aesthetic judgment to emotionally expressive Iranian classical dance stimuli. To explore a different affective response to Iranian classical dance than *aesthetic judgment* (provided by participants to each of the stimuli), we also asked our participants one final question at the end of the experiment about how important they found *Persian dance*[2] in general. Based on prior research, we added education, age, dance experience and aesthetic responsiveness as covariates to all our analyses.

Our central research questions were (i) whether there is a difference between cultures in aesthetic judgment and importance ratings to Persian dance, and (ii) whether self-ascribed culture group or enculturation measures are better predictors in such analyses.

Predictions

Predictions were pre-registered on the OSF. We investigated, first, whether Iranian participants' aesthetic judgment (liking ratings) of Iranian classical dance movements differed from that of English and Southeast Asian participants, as a function of five different emotions that are expressed in the dance movements of the Iranian classical dance movement library. We predicted a main effect of emotion, and that pair-wise comparisons would show that stimuli intended to express fear, anger, and neutrality would be liked less than stimuli expressive of joy and sadness, in accordance with previous literature (Christensen et al., 2025, Smith & Cross, 2022). Furthermore, Iranians' were expected to give higher (more positive) aesthetic judgments overall compared to those of English participants, regardless of intended emotion of the clips. We also expected an interaction effect between emotion and group on aesthetic judgment, and that pair-wise comparisons would show a difference in aesthetic judgment for stimuli intended to express different emotions (pre-registered). Second, with the aim of replicating previous research on the effects of a number of covariates, the following variables were added to the models: aesthetic responsivity, measured by means of the Aesthetic Responsiveness Assessment (AReA; in Farsi: Golbabaei et al., 2022; in English: Scholtz et al., 2020), dance experience, age, and education (pre-registered). Third, we investigated (i) which variable predicts aesthetic judgment best: (a) categorical culture group membership (Iranian/English/Southeast Asian participants as selected via the respective filters on the recruitment platform, Prolific®), or (b) three continuous questionnaire scores for enculturation with Iranian and English traditions and arts (questionnaires) (Christensen et al., 2024a, 2024b). We predicted that continuous enculturation measures would predict aesthetic judgment and importance of Persian dance, and that these would have a larger effect size than the categorical culture variable (pre-registered).

Finally, we conducted a series of exploratory analyses that had not been pre-registered. Given that a previous study with the same stimuli set reported an effect of *movement type* on aesthetic judgment (Khorsandi, et al., *under review*), we also included this variable in the analyses. While arm movements are a major focus of the expression, and hand movements punctuate the expressivity of the dancer's intention, adagio movements and turns (including any rotating movements) provide the flow that is integral to the aesthetic quality of this dance genre. Besides, lower body and feet produce an essential rhythmic foundation for the movements of the upper body. For instance, the lower body 'dogam' movement produces a visual 'pulse' that underlines the fluid arm gestures and intricate hand motions. The interplay between upper and lower body in this dance style is delicate and may remain unnoticed by non-Iranian observers. Thus, we explored whether this was also the case for participants in the preset study, predicting that, as before, turns would be liked most, lower body movements, least (not pre-registered). Similarly, previous studies have shown an effect of kinematic features like motion energy on aesthetic judgment (Christensen et al., 2016; Christensen et al., 2025; Khorsandi et al., *under review;* Orlandi et al., 2020), hence, we also added this variable as a covariate (not pre-registered). We predicted that higher motion energy would result in higher aesthetic judgment.

[2] When this experiment was set up, our team still referred to this dance style as 'Persian dance'. The term, in fact, refers to the same dance style as Iranian classical dance or Iranian Urban dance, as explained above. 'Persian' is also a widely used term to refer to other artforms from Iran (i.e. "Persian paintings"), in accordance with general nomenclature.

As a final exploratory analysis, unrelated to the aesthetic judgment of *liking* mentioned above, we investigated which variables (culture group, enculturation measures, aesthetic responsiveness, education, age and dance training) would predict how important a person finds Persian dance in general, assessed via Likert scale with an opinion-based question: *How Important Do You Find Persian Dance?* We hypothesised that Iranians would show the highest ratings (not pre-registered).

Method

The experiment was covered under the umbrella approval of the Max Planck Institute for Empirical Aesthetics (Nr. 2017_12). The experiment was performed online via the platform Prolific® (Peer et al., 2021; Stanton et al., 2022), and informed consent was obtained from all participants before participation began through an online tick-box system. All methods were performed in accordance with the relevant guidelines and regulations, and in accordance with the Declaration of Helsinki (World Medical Association, 1964).

This paper is part of a larger project from which different papers have emerged, examining research questions concerning cross-cultural emotion recognition and aesthetic judgment (Christensen et al., 2025; Christensen et al., 2024a, 2024b; Khorsandi et al., *under review*).

The hypotheses of this experiment were pre-registered on the OSF as part of a larger project: https://osf.io/ptg4x?view_only=4b84b60db61b4f7d9750827c10eb6070

Data were obtained in 2023. Data from this and related experiments are available on the OSF: https://osf.io/2evgt/?view_only=deb256becb1842adb154818b1f6c77de

Participants

A total of 120 participants took part in the experiment; 40 Iranian participants (20 male; age all: $M = 31.2$; SD $= 7.80$), 40 English participants (20 male; age all: $M = 32.6$; $SD = 8.94$), and 40 Southeast Asian participants (20 male; age all: $M = 31.5$; $SD = 8.99$). Considerable care was taken to match the three groups carefully by means of the Prolific® filters, with regards to study-relevant variables (age, sex, education). The Iranian participants were collected first because this was the smallest total group available on Prolific (approximately 200 participants in 2023). As discussed later, the Iranian participants were all individuals living in diaspora, as Prolific is not reachable from inside Iran. The filters for selecting the Iranian participants were set to 'country of birth' (Iran), 'first language' (Farsi). In addition, we had a demographics question that asked for participants' 'country of residence during 12 first years of life'. Once a sample of 40 Iranian participants had participated, demographic variables were inspected with regards to age, sex and education. Then filters were adjusted and the Southeast Asian participants were collected. For this group, besides the demographics filters (age, sex, education), the filters were set to 'country of birth' (England), 'first language' (any of the Southeast Asian languages, e.g., Urdu, Bengali, etc., see Table 1), 'current country of residence' (England). For the demographics question that asked for participants' 'country of residence during 12 first years of life', we checked the submissions and included only participants who indicated 'England'. Finally, the English participants were invited by again setting the filters accordingly for age, sex and education, and in addition the filters were set to 'country of birth' (England), 'first language' (English), 'current country of residence' (England). This means that the final set of 120 participants included Southeast Asian and English participants whose current country of residence was England, while the Iranian participants were currently living in any country except Iran (we did not collect their current country of residence). See Table 1 for participant characteristics.

Materials

Stimuli

Twenty-five stimuli were selected from the 100 stimuli of the Iranian classical dance movement library (6s length). The full stimuli library contains 20 sequences of movements danced five times each by a professional Iranian dancer. The dancer endowed her movements with a different emotional intention at each repetition (angry, fearful, joyful, sad, or neutral). The five stimuli with the highest recognition rate for each of the five emotion were selected (totalling 25

Table 1 Sample characteristics

Baseline characteristic		Iranian (living in diaspora)	English (in England)	Southeast Asian (in England)
Age		31.18 (7.80)	32.25 (8.94)	31.45 (8.99)
		range = 39	range = 31	range = 39
Gender	Female	19 (47.50%)	19 (47.50%)	20 (50.00%)
	Male	20 (50.00%)	20 (50.00%)	20 (50.00%)
	Prefer not to say	1 (2.50%)	1 (2.50%)	0 (0%)
Education	Did not complete school	0 (0%)	0 (0%)	0 (0%)
	Completed school up to age of 16 (e.g., GCSE)	0 (0%)	3 (7.50%)	1 (2.50%)
	Completed further education (e.g., A-level, diploma)	2 (5.00%)	5 (12.50%)	9 (22.50%)
	Some university or professional qualification	3 (7.50%)	5 (12.50%)	1 (2.50%)
	Completed undergraduate or professional qualification	12 (30.00%)	16 (40.00%)	21 (52.50%)
	Completed postgraduate degree (e.g., MA, PhD)	23 (57.50%)	11 (27.50%)	8 (20.00%)
	Prefer not to say	0 (0%)	0 (0%)	0 (0%)
Total years of education		18.00 (5.51)	16.75 (2.51)	16.95 (3.23)
		range = 25	range = 10	range = 25
Lived first 12 years of life	Iran	35 (87.50%)	0 (0%)	0 (0%)
	United Kingdom	2 (5.00%)	37 (92.50%)	40 (100%)
	Denmark	0 (0%)	1 (2.50%)	0 (0%)
	Estonia	0 (0%)	1 (2.50%)	0 (0%)
	Germany	0 (0%)	1 (2.50%)	0 (0%)
	Emirates	1 (2.50%)	0 (0%)	0 (0%)
	Netherlands	1 (2.50%)	0 (0%)	0 (0%)
	South Africa	1 (2.50%)	0 (0%)	0 (0%)
Mother tongue	Farsi	38 (95.00%)	0 (0%)	0 (0%)
	English	1 (2.50%)	37 (92.50%)	28 (70.00%)
	Bengali	0 (0%)	0 (0%)	3 (7.50%)
	Gujarati	0 (0%)	0 (0%)	4 (10.00%)
	Punjabi	0 (0%)	0 (0%)	3 (7.50%)
	Other	1 (2.50%)	3 (7.50%)	2 (5.00%)
Multi-lingualism	1 additional spoken language	27 (67.50%)	31 (77.50%)	35 (87.50%)
	2 additional spoken languages	9 (22.50%)	6 (15.00%)	4 (10.00%)
	3 or more additional spoken languages	4 (10.00%)	3 (7.50%)	1 (2.50%)
Languages spoken	Arabic	(0%)	2 (1.67%)	1 (0.83%)
	Bengali	(0%)	(0%)	5 (4.17%)
	Cantonese	(0%)	(0%)	4 (3.33%)
	English	4 (3.33%)	37 (30.83%)	12 (10.00%)
	German	4 (3.33%)	2 (1.67%)	1 (0.83%)
	Italian	4 (3.33%)	4 (3.33%)	(0%)
	Punjabi	(0%)	(0%)	5 (4.17%)
	Spanish	9 (7.50%)	(0%)	(0%)
	Welsh	5 (4.17%)	(0%)	(0%)

Table 1 continued

Baseline characteristic		Iranian (living in diaspora)	English (in England)	Southeast Asian (in England)
	Other	28 (23.33%)	13 (10.83%)	19 (15.83%)
Formal dance experience	Yes	6 (15.00%)	11 (27.50%)	1 (2.50%)
	No	34 (85.00%)	29 (72.50%)	39 (97.50%)
	Average years of dance experience	1.58 (6.30)	2.13 (4.22)	0.13 (0.79)
		range = 34	range = 17	range = 5
Dance styles	Ballet	1 (2.50%)	9 (22.50%)	2 (5.00%)
	Disco (free dancing in the club)	1 (2.50%)	3 (7.50%)	1 (2.50%)
	Hip hop and other street dance styles	0 (0%)	2 (5.00%)	1 (2.50%)
	Jazz dance	0 (0%)	3 (7.50%)	0 (0%)
	Modern dance/contemporary	1 (2.50%)	5 (12.50%)	0 (0%)
	Persian dance	13 (32.50%)	0 (0%)	0 (0%)
	Tap	0 (0%)	5 (12.50%)	0 (0%)
	Other	7 (17.50%)	2 (5.00%)	1 (2.50%)
Social dance enjoyment	Never	7 (17.50%)	5 (12.50%)	23 (57.50%)
	Two or three times per week	4 (10.00%)	2 (5.00%)	0 (0%)
	Once per week	3 (7.50%)	7 (17.50%)	2 (5.00%)
	Once per fortnight	5 (12.50%)	2 (5.00%)	1 (2.50%)
	Once per month	10 (25.00%)	6 (15.00%)	2 (5.00%)
	Every couple of months	2 (5.00%)	8 (20.00%)	5 (12.50%)
	A few times a year	9 (22.50%)	10 (25.00%)	7 (17.50%)
Persian dance	How important is Persian dance for you?	73.68 (20.85)	26.18 (20.94)	26.58 (26.45)
Task interest	How much did you enjoy this task?	3.43 (0.68)	3.67 (0.64)	3.42 (0.76)

N = 120. Mother tongue: Category "Other" summarizes languages that were referenced as a native language by less than two participants. The category includes Chinese, Danish, Dutch, Estonian, French, and Urdu. "Languages Spoken" includes languages spoken additionally to the native language. The category "Other" summarizes languages that are spoken by less than three participants and includes: Afrikaans, Catalan, Chinese, Czech, Danish, Dutch, Estonian, Farsi, French, Greek, Gujarati, Hebrew, Korean, Kurdish, Mandarin, Polish, Romanian, Russian, Swedish, Tamil, Thai, Turkish, and Urdu. Percentages are displayed as a fraction of the total sample size of $n = 120$. Dance styles: Category "Other" summarizes dance styles that were mentioned by two or less participants. Includes: African Dances, Argentine Tango, Belly Dance, Burlesque/Lyrical Dance, Folklore, Salsa/Bachata and Indian Dances. Please note that this table is the same as in (Christensen et al., 2025) for Experiment 2, as it was the same participants who provided both emotion recognition performance (Christensen et al., 2025) and aesthetic judgment (the present investigation)

stimuli), based on the emotion recognition values from Christensen et al. (2025) (Fig. 1).

The stimulus creation procedure is set out in detail in Christensen et al. (2025). Briefly, the dance sequences had been choreographed by the dancer so that there were roughly the same number of movements in each stimulus. There were four categories of stimuli in terms of the focus of the movement. These categories were (1) lower-body focus (e.g., dogam movements, or the triple step), (2) adagio movements (slow movements), (3) staccato movements (fast upbeat movements, often isolating parts of the body), and (4) turns (e.g., movements on the spot swirling

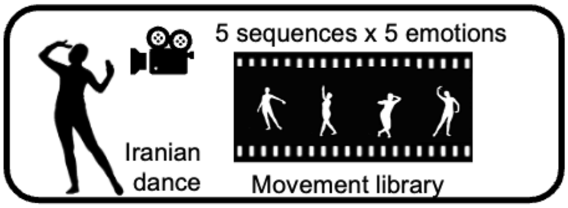

Fig. 1 Stimuli. Twenty-five stimuli (6s length) selected from Christensen et al. (2025). Stimuli were recorded with a greenscreen, leaving only a white silhouette visible against a black background

around, or turning across the floor). The selected stimulus set of 25 included the five emotion categories (anger, fear, sadness, joy, neutrality) and contained instances from the four movement types (adagio, lower body focus, staccato, and turns). We use both categorizations (emotion and movement type) as categorical variables in our analyses.

Motion energy for each stimulus was obtained via a customized Matlab script (from Christensen et al., 2016). The script first converts the video into greyscale and counts any pixel change in luminance for each frame and then computes the number of pixels that changed in luminance from frame n to frame n + 1. The frame information is summed up over the video frames (for each video) and then averaged to obtain the final value for each video.

Questionnaires

Responses to three validated enculturation screening tools, along with an aesthetic responsiveness measure, and demographics questions were collected in Farsi and English.

The aesthetic responsiveness assessment (AReA) The Aesthetic Responsiveness Assessment has 14 items and answers are given on a 5-point Likert scale between 0 ("never") and 4 ("very often"). The questions probe for the frequency with which participants go to cultural events and locations, whether they feel strongly when they engage with the arts and whether they practice any arts themselves. Three subscales gauge these three domains (locations, feelings, practices) of an individuals' engagement with different types of artistic materials. Only the total score was used within this investigation. The reliability of the scales range from α = .63 to .84 in English (Schlotz et al., 2020), and between α = .72 to .78 in Farsi (Golbabaei et al., 2022), suggesting good reliability.

The cultural traditions questionnaire (CTQ) The Cultural Traditions Questionnaire (CTQ) measures enculturation with Iranian and English cultural traditions: Christmas and Easter in England, and Yalda and Norouz in Iran. There are 16 items about Cultural Traditions; eight about England, eight about Iran (CTQ-en/ir). The questions are about actual practice (there are no knowledge-based questions), and include, for example, "*How intensely do you celebrate Nowruz? (e.g. do you have family dinners, take time off, give or receive Eidy or presents?)*" or "*How much effort did your parents put into making Nowruz a beautiful experience every year for you when you were under 12 years old?*". Responses are given on a 6-point rating scale from 1 ("not at all") to 6 ("very much"). Reliabilities for the two scales in each language range between α = .86 to .97 (Christensen et al., 2024a, 2024b).

The arts engagement in childhood questionnaire (AECQ) The Arts Engagement in Childhood Questionnaire (AECQ) probes for the level of exposure to Iranian and English arts (dance, music, paintings) from a specific time period (English Impressionism and Qajar period respectively; approx. 1850–1920), during the individuals' sensitive enculturation period, before the age of 12 years. Questions target specific experiences. For example: "Did you see Persian miniature paintings in your home, in your every day life settings, like at school, in libraries, etc? Was it something that people had in their homes, that art teachers would speak about, your parents would know about …?" or "Did you see Persian Dance on television and in print magazines when you were little?" There are 18 items; 9 items about English, 9 items about Iranian arts engagement. Responses are given on a 6-point rating scale from 1 ("not at all") to 6 ("very much"). Reliabilities for the two scales in each language range between α = .75 to .91 (Christensen et al., 2024a, 2024b).

The enculturation and acculturation quiz (EAQ) The Enculturation and Acculturation Quiz (EAQ) examines individuals' enculturation via *knowledge* about specific cultural practice and their cues. Participants choose via binary answers (yes/no) among a list of items, whether a given item is part of a specific cultural practice (Christmas, Easter, Yalda, and Norouz). For example, for English Christmas traditions, the item "fir tree" is a "correct" answer, while "chocolate egg" is not (Christensen et al., 2024a, 2024b). The participant has to know whether an item is part of the tradition in question (these are all knowledge-based questions and do not probe for experience with the tradition).

For scoring, the quiz uses the F1 measure (from information retrieval theory), which is an accuracy measure that tries to balance all types of responses to measure the amount of relevant information. It is the harmonic mean between precision, TP/(TP+FP), and recall, TP/(TP+FN), i.e. F1 = 2 * TP/(2 * TP + FP + FN), where TP = true positives, FP = false positives, and FN = false negatives. In this measure, true negatives (TN) are disregarded due to the task type, because not selecting anything would result in an accuracy of 50%, and bias the accuracy. The measure also gives penalties to missed correct items as well as to falsely recognized items. The value range is between 0 (no correct answer) and 1 (only correct answers). Split-half reliability (over odd and even numbered items) showed very high reliabilities; for the full set: $r = .89$; for English only $r = .89$, and for or Iranian only $r = .83$.

Procedure

Limesurvey© was used to set up the experiment in English and in Farsi. Participants performed the task, and were paid for their participation through the Prolific® online behavioral experimentation platform (Peer et al., 2021). During the study advertisement process (powered by Prolific), it was made clear that the survey had to be done on a desktop computer and not with a phone or tablet. We did not obtain further information about the screen sizes of the participants. A question was included at the beginning of the survey to ensure that participants would do the experiment with Google Chrome or Firefox because only these two browsers at the time of the experiment allowed the autoplay function for the videos to play.

The survey had two parts. Part 1 was the aesthetic judgment task, where participants watched 25 video stimuli (6s each), in random order, one by one. After each presentation, participants made their aesthetic judgment of *liking* on a Likert slider-scale from 0 ("not at all") to 100 ("very much"). At the very end of the experiment, participants were asked *once*, 'How important is Persian dance for you?'. Part 2 of each survey included the demographics questions, the AReA and the enculturation questionnaires (Fig. 2).

Results

Before conducting our pre-registered models, we checked for normality and found that the aesthetic judgment variable liking and motion energy were not normally distributed, nor were the questionnaire measures. Shapiro-Wilk tests for normality showed that normality had been violated for all variables (all *p*s $< .001$). Therefore, we produced z-scored values of these variables for our further analyses.

Next, we produced bi-variate Pearson correlations between all variables of interest. We found too high intercorrelations between the six scales of the three enculturation variables (AECQ, CTQ, and EAQ for both cultures, English and Iranian). Pearson's correlation coefficients were between .5 and .9. See Table 2 for the correlation matrix.

In such cases, it is customary to produce average scores of the z-scores of the intercorrelated variables, which we did. This resulted in two new variables: the Composite Culture Score for English culture (CCS-en = (zAECQ-en + zCTQ-en + zEAQ-en)/3), and the Composite Culture Score for Iranian culture (CCS-ir = (zAECQ- ir + zCTQ- ir + zEAQ- ir)/3). These two variables were used instead of the six separate values in the subsequent ANOVAs.

A Multivariate ANOVA was conducted with the two variables CCS-en and CCS-ir as dependent variables and culture group as between-groups factor. The resulting model was significant for both CCS-en ($F(2, 117) = 23.489$, $p < .001$, partial $\eta^2 = .286$) and CCS-ir ($F(2, 117) = 569.572$, $p < .001$, partial $\eta^2 = .907$). Šidák-adjusted pairwise comparisons indicated that English participants had the highest scores on the CCS-en, followed by the Southeast Asian participants ($p < .001$). Iranian participants also scored lower on the CCS-en, than English participants ($p < .001$), while there was no difference between Iranian and Southeast Asian participants on the CCS-en ($p = .504$). Iranian participants scored higher on the CCS-ir, than English ($p < .001$), and Southeast Asian participants ($p < .001$), while the English participants and Southeast Asian participants did not differ on the CCS-ir ($p = .946$). These results are set out in Fig. 3 and show that the three groups differed from each other on key

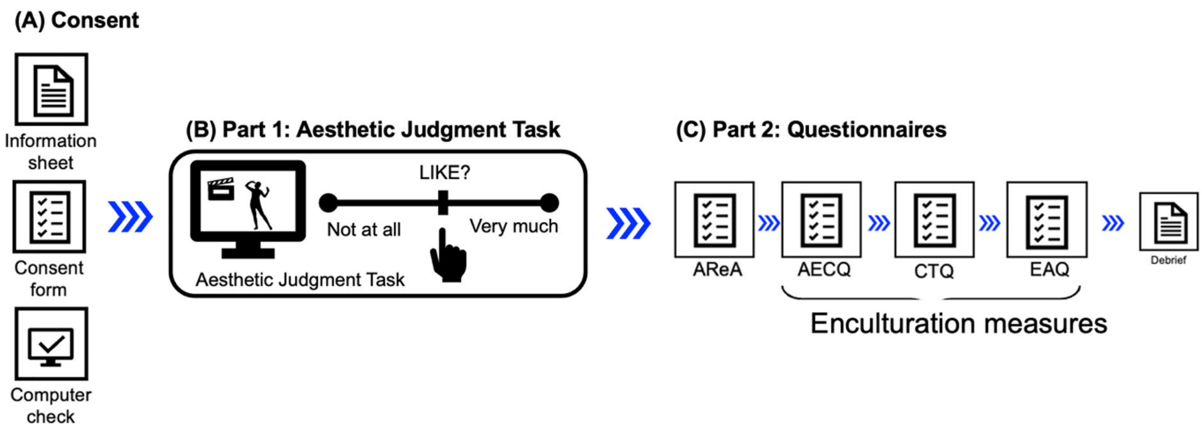

Fig. 2 Procedure. **A** Participants performed the task online and provided informed consent via a tick-box click. A browser question ensured that participants were performing the task on a desktop computer (not on phone or tablet) and with the browsers Chrome or Firefox (to ensure seamless running of the videos). **B** Part 1 was the aesthetic judgment task in which participants rated each stimulus in terms of how much they liked it (0 = not at all; 100 = very much). Part 1 ended with a single question with a Likert slider-scale 'How important is Persian dance for you?' (0 = not at all; 100 = very much). The importance question was only asked *once*. **C** Part 2 contained all questionnaire measures and ended with a debrief page

enculturation scores for both English and Iranian culture. English participants were more enculturated to English culture than Iranian and Southeast Asian participants, while Iranians were more enculturated with Iranian culture than both English and Southeast Asian participants.

Results: impact of culture, enculturation, and *intended emotion* on aesthetic judgment

A mixed linear model was conducted, with the within-group factor intended emotion (anger, fear, joy, neutral, sad), the between group factor culture (Iranian, English), and the covariates motion energy, age, education (years), AReA, CCS-en, and CCS-ir. SubjectID and stimuli were random effects variables. The dependent variable was participants' aesthetic responses (Liking judgment, z-scored; 0 = "not at all", 100 = "very much").

There were main effects of intended emotion ($F(4, 19) = 6.84$, $p < .001$, partial $\eta^2 = .590$), of the AReA ($F(1, 112) = 5.26$, $p = .024$, partial $\eta^2 = .045$), and of CCS-en ($F(1, 112) = 6.016$, $p = .016$, partial $\eta^2 = .051$). There were no main effects of motion energy ($p = .800$, partial $\eta^2 = .003$), education ($p = .522$, partial $\eta^2 = .004$), age ($p = .248$, partial $\eta^2 = .011$), culture ($p = .259$, partial $\eta^2 = .024$) and CCS-ir ($p = .641$, partial $\eta^2 = .002$), nor an interaction between intended emotion and culture ($p = .219$, partial $\eta^2 = .004$).

Šidák-adjusted pair-wise comparisons revealed that the main effect of intended emotion meant that joyful movements were liked most of all, followed by angry, fearful, sad and neutral movements. There were significant differences between joyful and neutral ($p < .001$), joyful and sad ($p < .001$), and between angry and neutral ($p = .033$) and between angry and sad ($p = .007$). See Fig. 4A.

There was no significant main effect of culture. Following our pre-registered hypotheses, we explored the comparisons nonetheless. Šidák-adjusted pair-wise comparisons revealed no significant differences between either of the groups (all $ps > .272$). See Fig. 4B.

The interaction intended emotion × culture was not significant; however, we had pre-registered the interaction, and, therefore, inspected the Šidák-adjusted pair-wise comparisons. Descriptively, the direction of the effect was the same across all cultural groups: joyful movements were liked above all, followed by angry, fearful, neutral, and sad. For the Iranian group there were significant differences between joyful and neutral and between joyful and sad movements (all $ps < .001$), and between sad and angry movements ($p < .05$). For the English participants the same differences were significant (all $ps < .05$). For the Southeast Asian participants, only the differences between anger and sadness and between joy and sadness were significant (all $ps < .05$). There were no significant differences

Table 2 Intercorrelations

	1. Liking (z-score)	2. Dance training	3. Motion energy (z-score)	4. Age	5. Education (years)	6. AReA score	7. AECQ-en (z-score)	8. AECQ-ir (z-score)	9. CTQ-en (z-score)	10. CTQ-ir (z-score)	11. F1-en (z-score)	12. F1-ir (z-score)	13. CCS-en (z-score)	14. CCS-ir (z-score)	15. Importance Persian Dance
1.	1														
2.	.059**	1													
3.	.054**	1	1												
4.	.090**	− .034	1	1											
5.	.041*	.136**	1	.207**	1										
6.	.104**	.174**	1	.120**	.266**	1									
7.	.133**	.511**	1	.158**	.156**	.250**	1								
8.	− .019	.141**	1	− .033	− .002	.355**	.045*	1							
9.	.129**	.160**	1	.104**	.050**	.231**	.397**	− .037*	1						
10.	− .089**	.038*	1	.009	.155**	.203**	− .078**	.715**	− .108**	1					
11.	− .069**	.071**	1	− .010	.156**	.197**	− .129**	.767**	.370**	.909**	1				
12.	.065**	.189**	1	.141**	− .051	− .035	.299**	− .348**	− .119**	− .511**	− .450**	1			
13.	.144**	.379**	1	.178**	.069**	.197**	.749**	− .150	.780**	− .331**	− .730**	− .286**	1		
14.	− .063**	.090**	1	− .012	.111**	.271**	− .058**	.890**	− .095**	.959**	− .470**	.940**	− .275**	1	
15.	.056**	.142**	1	.092**	.097**	.357**	− .021	.621**	.068**	.727**	− .435**	.607**	− .172**	.701**	1

CCS-en ComposiTE CULTURE SCore England, *CCS-ir* Composite culture score Iran. Pearson's correlation coefficient for aesthetic judgment (liking; 0 = "not at all", 100 = "very much") with variables of interest of N = 120 participants (40 English, 40 Iranian, 40 Southeast Asian). *p <.05 **p < .001

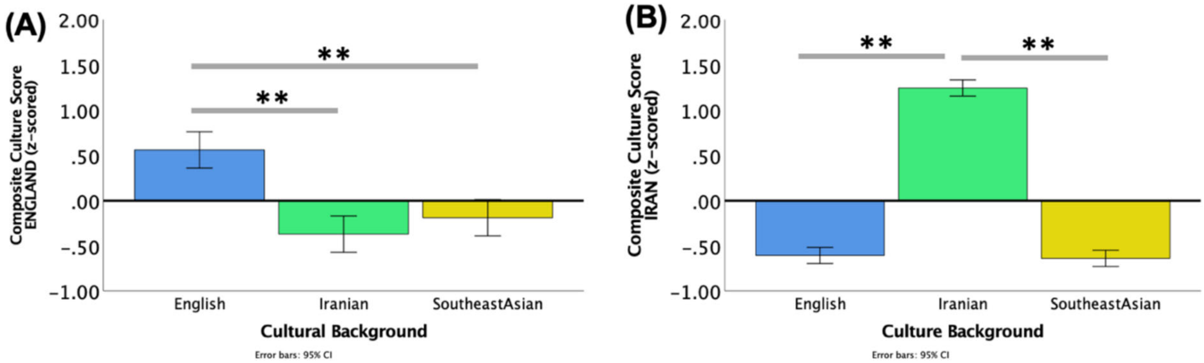

Fig. 3 Main effect of culture on composite culture scores for England and Iran (z-scored). **A** Main effect of Cultural Background showing the three cultural groups and their z-scored Composite Culture Score for English culture (CCS-en = (zAECQ-en + zCTQ-en + zEAQ-en)/3). **B** Main effect of Cultural Background showing the three cultural groups and their z-scored Composite Culture Score for Iranian culture (CCS-ir = (zAECQ-ir + zCTQ-ir + zEAQ-ir)/3). Error bars: 95% CI. $^{**}p < .001$.

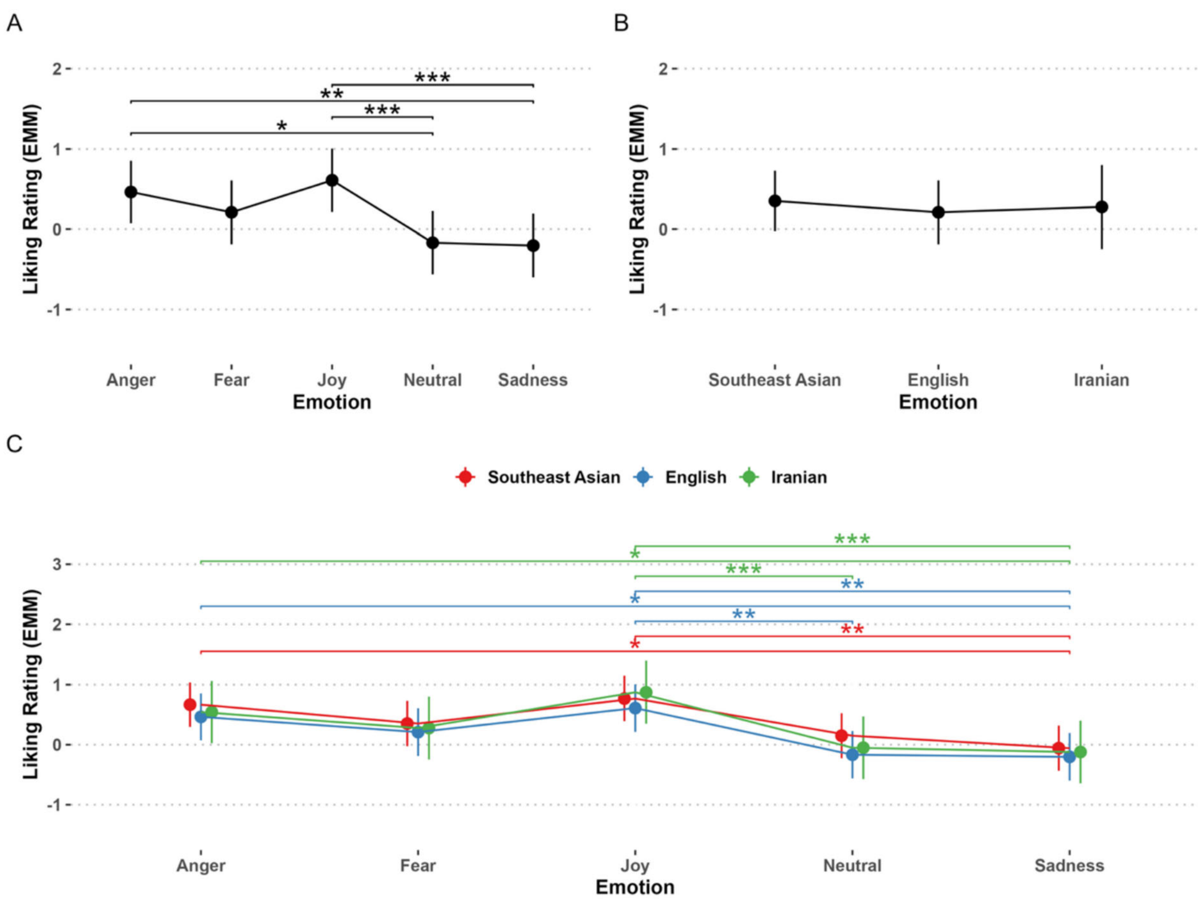

Fig. 4 Main effects and interactions of Intended Emotion and Culture. Main Effects of Emotion, Movement Type and Cultural background on aesthetic judgment (liking ratings; 0 = 100, z-scored Estimated Marginal Means, EMM) **A** Main Effect of Intended Emotion. **B** Main Effect of Culture. **C** Interaction Emotion × Culture. Covariates appearing in the model are evaluated at the following values: Motion Energy = 8654.83, education (years) = 17.23, dance training (years) = 1.28, age = 31.63, AReA = 25.71, CCS-en = 0.00000, CCS-ir = 0.00000. $^{*}p < .05$ $^{**}p < .008$

between the groups between the emotions. See Fig. 4C.

Results: impact of culture, enculturation, *movement type* on aesthetic judgment (not pre-registered)

For exploratory purposes, a mixed linear model was conducted, with the within-group factor movement type (adagio, lower body, staccato, turn), the between group factor culture (Iranian, English), and the covariates motion energy, age, education (years), AReA, CCS-en, CCS-ir. SubjectID and stimuli were random effects variables. The dependent variable was participants' aesthetic responses (Liking judgment, z-scored; 0 = not at all, 100 = very much).

There were main effects of movement type ($F_F(3, 20) = 13.613$, $p < .001$, partial $\eta^2 = .671$), of the AReA ($F(1, 112) = 5.257$, $p = .024$, partial $\eta^2 = .045$), and of CCS-en ($F(1, 112) = 6.017$, $p = .016$, partial $\eta^2 = .051$). There were no main effects of of culture ($p = .450$), of age ($p = .248$), motion energy ($p = .638$), education ($p = .522$), CCS-ir ($p = .641$), nor an interaction between mode and culture ($p = .306$).

Šidák-adjusted pair-wise comparisons revealed that there were significant differences between all movement types (all ps $< .001$, except staccato-adagio $p = .156$), in that the turns were liked most of all, followed by adagio, and staccato, while lower body movements were liked the least. In Fig. 5, we have plotted the interaction movement type x culture (even if not significant) to show how the pattern of aesthetic judgment is very similar across cultural groups and mainly driven by the movement type.

Results: impact of culture, enculturation and demographics on importance of *Persian dance* (not pre-registered)

A significant Oneway-ANOVA (F(2, 117) = 56.900, $p < .001$, $\eta^2 = .493$), and Šidák-corrected pairwise comparisons showed that Iranians found Persian dance more important than English and Southeast Asian participants (ps $< .001$) who did not differ between each other ($p = 1$).

We next modelled participants' ratings of how important Persian dance is to them with a multiple linear regression (dependent variable = importance, range; 0 = not at all to 100 = very much, *z-scored*). As predictors we entered culture, education (years), dance training (years), AReA and the CCS-en and CCS-ir.

We continued using z-scored variables, also for the importance of Persian dance variable. Besides, because the culture questionnaires were highly inter-correlated and a preliminary regression showed that these scores had high variance inflation, we again used the averaged z-scored culture questionnaire scores (CCS-en/ir), and included these instead of the separate culture questionnaire scores.

A significant regression was found ($F(7, 119) = 18.72$, $p < .001$), with an R^2 of .54, meaning that this model explained 54% of the variance in the variable 'importance of Persian dance' with seven predictors. Of the variables entered into the model, two were significant; the AReA ($t = 2.64$, $\beta = .189$, $p = .010$), and the CCS-ir ($t = 8.53$, $\beta = .626$, $p < .001$). The other predictors were not significant: Culture group ($t = .823$, $\beta = .059$, $p = .412$), age ($t = 1.387$, $\beta = .094$, $p = .168$), dance training ($t = 1.266$, $\beta = .091$, $p = .208$), education ($t = .037$, $\beta = .002$, $p = .971$), nor was the CCS-en ($t = -.064$, $\beta = -.773$, $p = .441$). See Table 3 for these results.

We inspected the variance inflation factor (VIF) for all our predictors in the model. None of our variables had a VIF of over the problematic value of 10 (Field, 2009), in fact the highest was only 1.6 (CCS-en). Next, we inspected the Condition Index which should ideally be under 10. For CCS-en it was 10.35 and for CCS-ir it was 16.86, which could suggest a small multicollinearity issue. However, when checking the variance proportions, none of the variables had values that were anywhere close to the problematic .9 value (all variance proportions were between .01 and .47). We therefore assume multicollinearity was not an issue.

Discussion and conclusion

We first tested the influence of emotion, culture, and enculturation on aesthetic judgments of Iranian dance sequences. Contrary to our predictions, Iranian did not like the movements more than the other cultural groups, in fact there were no differences between any

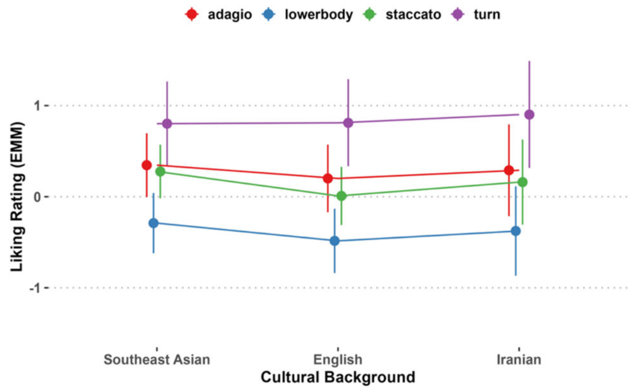

Fig. 5 Movement type by culture interaction. Z-scored estimated marginal means (EMM) of liking judgments, illustrated as a function of culture group. The interaction was not significant (p = .509). We plotted it to show, how the pattern of aesthetic judgment is very similar across cultural groups and mainly driven by the movement type. There were significant differences between all movement types (all ps < .001, except staccato-adagio p = .022), in that the turns were liked most of all, followed by adagio, staccato and lower body movements which were liked least. Covariates appearing in the model are evaluated at the following values: Motion Energy = 8654.83, education (years) = 17.23, dance training = 1.28, age = 31.63, area = 25.71, CCS-en = .00000, CCS-ir = 0.00000. Error bars: 95% CI

Table 3 Regression table for importance of Persian dance

DV: Importance of Persian dance (z-scored: original 0 = not important; 100 = very important)					
	B	*SE B*	β	*t*	*p*
Model					
Constant				− 2.328	.022*
Cultural background (prolific filter)	.073	.088	.059	.823	.412
Education (years)	.001	.017	.002	.037	.971
Age	.011	.008	.094	1.387	.168
Dance experience (years)	.021	.016	.091	1.266	.208
AReA score	.018	.007	.189	2.636	.010*
CCS-en	− .085	.110	− .064	− .773	.441
CCS-ir	.673	.079	.626	8.529	< .001**

B Beta, β Standardized beta.
*$p < .05$, **$p < .001$

of the groups. We replicated previous results in that turn movements were liked most of all, followed by adagio, staccato, and lower body movements. However, contrary to previous results (Khorsandi, et al., *under review*), here we found no effect of culture on movement type, and motion energy was also not a significant factor for aesthetic judgment, as has been reported in previous studies (Christensen et al., 2016, 2019, 2021; Khorsandi et al., *under review*).

Joyful movements were recognized most in the validation study (Christensen et al., 2025), meaning that in a forced-choice emotion recognition task, both Iranian and English participants had recognized movements where the dancer intended to express joy over and above all other emotions (followed by angry, sad, and fearful movements). There were significant differences in emotion recognition scores between emotions, except between neutral and sad movements. Here, liking judgments followed the same order descriptively, and joyful and angry movements were liked significantly more than sad and neutral movements. Inspecting pre-registered pair-wise differences in the interaction between intended emotion and culture, we found a slightly more differentiated pattern of results between the three culture groups. who showed significant differences in their liking ratings for all pairs (except between sad-neutral). Iranians and English participants had the most fine-grained aesthetic judgment, and were especially sensitive to the joyful emotional expressivity of the dancer (the pairs joy-sadness and joy-neutral were significant). Conversely, Southeast Asian participants did not distinguish in their aesthetic judgment between joyful and neutral movements. We can only speculate about the reason for this. However, based on the fact that joyful dance is a stylistic norm in Iran, perhaps the difference between joyful, sad and neutral expressivities was starker for Iranians, and they therefore prefered joyful ones, based on their familiarity for this expression in dance.

Second, inspecting effect sizes, we found that the enculturation measures explained more of the variance than the cultural group variable. The categorical culture variable had a lower effect size, than the continuous CCS-en variable. Interestingly, enculturation with English culture was a stronger predictor of participants' aesthetic judgment, than enculturation with Iranian culture was. The latter was not significant, and the effect size much smaller. This could perhaps be explained by the fact that most of the stimuli were intended to express neutrality or negative emotions (20 out of 25 stimuli). Expressing negativity in this Iranian dance style is unusual, while in many Western dance styles, expressing sadness, fear, or even anger is more commonplace. These results offer a first insight into the usefulness of using continuous measures. Of course, such continuous measures require longer participant time, and more complex analyses, while categorical culture demographics are much more widely used in psychology and neuroscience. However, their operationalization of categorical culture demographics is often incoherent or based on unjustified grounds (with diverse proxies in usage, including world region, country, nation, society, ethnicity). This is problematic as it can make it difficult to interpret small effect sizes or null results of expected cultural effects. This paper, therefore, proposes that continuous enculturation measures, even if more laborious to design and longer to administer, will offer more granular insights into enculturation effects on aesthetic judgment. For example, the demonstration that English enculturation (but not Iranian enculturation) predicted aesthetic judgments, while Iranian enculturation predicted importance ratings, provides insights into the multidimensional nature of cultural influence on perceptual and cognitive response—while the categorical culture variable had returned a null result.

Despite showing no difference from the other cultural groups on aesthetic judgment of *liking,* Iranians' opinions about the Importance of Persian Dance were the highest. Persian dance was more important for Iranians, than for English and Southeast Asian participants (both ps $< .001$), who did not differ from each other ($p = 1$). Interestingly, a multiple regression showed that the enculturation measure with Iranian culture (CCS-ir) was a stronger predictor ($p <$ .001) than the culture variable (which was no longer significant within the model). Enculturation with Iranian culture also explained the variance in the rating much more strongly, than enculturation with English culture (which was not significant). The standardized beta for the continuous CCS-ir variable was higher (.63), than the beta for the categorical culture variable (.06). This importance rating was additionally only modulated by the AReA, but not by age, nor education.

Besides enculturation with Iranian culture, the aesthetic responsiveness of participants (assessed by means of the AReA: Schlotz et al., 2020), regardless of culture group, was a strong predictor for both aesthetic judgment and for the importance of Persian dance. Measuring such inter-individual differences has previously been shown to be important in empirical aesthetics research for explaining response variability (Schlotz et al., 2020), and our study confirms this once more.

We must discuss the possibility that the dual finding that Iranian participants showed no difference in aesthetic ratings to our stimuli, but the highest importance ratings to the question about the importance of Persian dance could be a sign of a flaw in our design. The stimulus set of Iranian dance was designed with the Western conceptualization of five universal emotions in mind (Christensen et al., 2025). As explained in Christensen et al. (2025) and above, for Iranian observers, seeing this dance style expressive of anything other than joy might have been unusual, violating its stylistic rules of expressivity being festive, positive, and generally joyful (Christensen et al., 2024a, 2024b). Besides, empirical assessments have since shown that the Iranian cultural sphere has different verbal labels for emotions perceived in the context of the arts (Borhani et al., 2025; Golbabaei et al., 2022). Our CCS scores may, thus, also have served as a cultural barometer of expertise with Iranian classical dance, the Iranians being more expert in this dance genre and therefore, also more critical. Expertise or familiarity has previously been shown to either increase or decrease observers' aesthetic judgments of dance movements and other artistic artifacts. Observers who are not familiar with the artworks generally like simple/less complex works, while experts prefer more complex ones, yielding an inverse-U-shaped relation between familiarity/expertise and aesthetic experience (Kirsch, et al. 2013; Jacobsen & Beudt, 2017; Song, et al. 2021). Thus, the higher expertise (or familiarity) of the Iranian participants in this study may have resulted in a more critical evaluation of the dance choreographies presented to them. Iranians were not only more familiar with the dance style, but also had slightly higher dance experience overall than the other groups, perhaps contributing to this more critical stance.

Our finding that continuous enculturation measures yielded higher effect sizes than categorical culture variables, merits further investigation. At the very least, these results suggest that continuous enculturation variables might be more useful than large-scale group membership, operationalized via proxies such as nation, country, society, or ethnicity for yielding statistically measurable insights into the role of culture for human perception and cognition.

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Author contributions JFC: Conceptualization, Methodology, Software, Validation, Formal analysis, Funding acquision, Project administration, Investigation, Resources, Data Curation, Writing—original draft, Writing—review and editing, Visualization, Supervision, Project administration, Funding acquisition. SK: Conceptualization, Resources, Data Curation, Formal analysis, Investigation, Methodology, Validation, Writing—review and editing. FU: Investigation, Resources, Visualization, Supervision, Project administration, Funding acquisition, Writing—review and editing. ESC: Investigation, Methodology, Validation, Supervision, Writing—review and editing. KF: Methodology, Formal analysis, Software, Validation, Data Curation, Visualization, Writing—review and editing. MV: Conceptualization, Data Curation, Project administration, Writing—review and editing. MW-F: Conceptualization, Funding acquision, Investigation, Resources, Visualization, Supervision, Writing—review and editing.

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Data availability Data from the Experiment are available at: https://osf.io/2evgt/.

Declarations

Competing interest The authors have no conflict of interest to disclose. Shahrzad Khorsandi owns a commercial dance school and dance company, however, the remaining authors do not see this as a conflict of interest.

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